ECA18 – OPTOELECTRONICS AND OPTICAL COMMUNICATION
CO2 PROBLEM SOLVING ASSIGNMENT

Standards-Compliant Laser Transmitter Design, Wavelength Control and Nonlinearity Management for an 8-Channel DWDM Link

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Course Outcome – CO2	Analyze the operation and principles of optical sources including LEDs and laser diodes (FP, DBR, DFB, VCSEL, QD), along with their characteristics such as modulation dynamics, spectral stability, efficiency, and the impact of nonlinear effects (SPM, XPM, FWM, SBS, SRS), soliton transmission, and photonic integration on system performance.
Nature of Assessment	Problem Solving Assignment
Duration	1 Week
Maximum Marks	100
Primary Bloom's Levels	K3 – Apply | K4 – Analyze

THE PROBLEM SOLVING ENGINEERING SITUATION

A communication system designer must develop an optical transmitter for an 8-channel DWDM fiber-optic link operating around the 1550 nm window with 100 GHz channel spacing. Each channel carries 25 Gb/s NRZ data over six amplified 80 km spans of single-mode fiber. The transmitter must provide adequate output power, high modulation speed, narrow linewidth, stable wavelength under ambient-temperature variation, and compliance with optical-interface standards and eye-safety limits. Candidate sources include an LED, Fabry–Perot (FP), DBR, DFB, VCSEL and quantum-dot (QD) lasers. The designer must then evaluate the nonlinear impairments (SPM, XPM, FWM, SBS and SRS), decide whether soliton transmission or digital backpropagation is appropriate, and select a discrete or photonic-integrated (InP or Si-photonics) transmitter implementation.

The proposed problem-solving study must satisfy the following information and constraints:

Parameter	**Requirement / Given Condition**
WDM configuration	8-channel DWDM link, 100 GHz spacing (≈ 0.8 nm), around 1550 nm
Link configuration	Six amplified spans, each 80 km of standard single-mode fiber
Candidate optical sources	LED, FP laser, DBR laser, DFB laser, VCSEL and QD laser
Target data rate	25 Gb/s NRZ per channel; approximate NRZ bandwidth relation $B \approx 0.7R_b$
Candidate laser data	$E_g = 0.80$ eV; radiative lifetime $\tau_r = 10$ ns; non-radiative lifetime $\tau_{nr} = 40$ ns; 3 dB modulation bandwidth = 20 GHz
Laser operating data	$I_{th} = 12$ mA; slope efficiency $\eta_s = 0.3$ mW/mA; bias current $I = 40$ mA; $P = \eta_s(I - I_{th})$
Temperature behaviour	Wavelength drift 0.1 nm/°C; ambient variation 40 °C; required wavelength tolerance ±0.1 nm
Fiber and launch data	$\alpha = 0.2$ dB/km; $\gamma = 1.3$ $W^{-1}km^{-1}$; $A_{eff} = 80$ μm²; launch power 2 mW per channel
Scattering data	$g_B = 5 \times 10^{-11}$ m/W; $g_R = 6 \times 10^{-14}$ m/W; $P_{th(SBS)} \approx 21A_{eff}/(g_B L_{eff})$; $P_{th(SRS)} \approx 16A_{eff}/(g_R L_{eff})$
Soliton data	$T_0 = 15$ ps; $\lvert\beta_2\rvert = 20$ ps²/km; $L_D = T_0^2/\lvert\beta_2\rvert$; $L_{NL} = 1/(\gamma P_0)$; soliton order $N^2 = L_D/L_{NL}$

STAGE 1 – INTERPRET THE COMMUNICATION REQUIREMENT

Work backward from the 8-channel, 25 Gb/s, 1550 nm DWDM requirement before beginning detailed source and nonlinear-transmission calculations.

Determine and discuss:

• the reason a suitable optical source must provide high modulation speed, narrow linewidth and high output power for the specified link;

• the source characteristics (bandgap, quantum efficiency, threshold, slope efficiency, linewidth) that determine wavelength, efficiency and spectral purity;

• the modulation, chirp and temperature-related quantities that must be controlled for reliable high-speed operation;

• the nonlinear-transmission quantities that become important when multiple DWDM channels, multiple spans and launch power are considered;

• the role of ITU-T G.957, IEEE 802.3bs/bm and ANSI Z136 eye-safety requirements in defining the transmitter specification;

• the system-level parameters that influence soliton or DSP-based compensation and the final photonic-integration decision.

Based on this analysis, identify the characteristics that an acceptable transmitter solution should possess.

SOLUTION – STAGE 1

Assumptions used in this solution

The question does not give a few values that are needed, so I assumed them. (1) Standard single-mode fibre has D = 17 ps/(nm·km) at 1550 nm; this is only used for walk-off, FWM and dispersion-tolerance estimates. (2) 1 Np = 4.343 dB, so $\alpha = 0.2$ dB/km = 0.0461 km^{-1}. (3) The eight channels sit on the ITU-T G.694.1 100 GHz grid, 193.1 THz to 193.8 THz (about 1552.5 nm to 1547.0 nm). (4) Every in-line amplifier restores the power to 2 mW per channel at the start of each span, and the six spans are identical. (5) The external modulator loses 5 dB and the AWG multiplexer loses 3.5 dB. (6) The TEC set point is 25 °C. (7) The eye-safety limit is taken as +10 dBm (10 mW) at 1550 nm, which is a conservative Class 1 type value; the exact figure should be confirmed from the current edition of the standard. (8) Values in the comparison tables (power, linewidth, speed) are typical datasheet or textbook values, not data from the question.

(a) Why the source needs high speed, narrow linewidth and high power

Each channel runs at 25 Gb/s, so one bit lasts $T_b = 1/(25 \times 10^9) = 40$ ps and NRZ needs $B \approx 0.7R_b = 0.7 \times 25 = 17.5$ GHz. The source (or its modulator) must follow this, otherwise the eye closes. A narrow linewidth is needed because the channels are only 0.8 nm apart, so a wide or multimode spectrum spills into the next slot, and because dispersion spreading goes as $D \cdot L \cdot \Delta\lambda$ over 480 km. Power is needed because each 80 km span loses $0.2 \times 80 = 16$ dB, so the OSNR falls span after span. But power is limited by the nonlinear effects and eye safety, and this trade-off runs through the whole assignment.

(b) Source characteristics that fix wavelength, efficiency and spectral purity

The bandgap sets the wavelength, $\lambda(\mu m) \approx 1.24/E_g(eV)$, so it must give 1.55 µm. The internal quantum efficiency $\eta_{int} = \tau_{nr}/(\tau_r + \tau_{nr})$ shows how much of the injected current becomes light instead of heat. The threshold current I_{th} is where gain equals loss, and the slope efficiency η_s gives $P = \eta_s(I - I_{th})$. The linewidth comes from spontaneous-emission noise (widened by $1 + \alpha_H^2$) and decides the spectral purity, the SBS threshold and the dispersion penalty.

(c) Modulation, chirp and temperature quantities

The things to control are the 3 dB bandwidth and the relaxation-oscillation frequency f_r that limits it, the bias point compared with threshold, and the extinction ratio. Chirp comes from the linewidth enhancement factor α_H, because the refractive index changes with carrier density when the current is modulated. Temperature moves the wavelength (0.1 nm/°C) and the threshold current, so with 0.8 nm spacing a TEC and a wavelength locker are needed.

(d) Nonlinear quantities

L_eff, A_eff, γ and the launch power give the nonlinear phase φ_NL = γPL_eff. The channel spacing and the dispersion D decide how strongly XPM and FWM act, g_B and g_R give the SBS and SRS thresholds, and everything adds up over the six spans.

(e) Role of G.957, IEEE 802.3bs/bm and ANSI Z136

ITU-T G.957 sets the transmitter limits for mean launched power, centre wavelength, spectral width, SMSR (30 dB minimum), extinction ratio and eye mask. It only goes up to STM-16, so I also use G.694.1 and G.959.1 for the grid and the DWDM interface. IEEE 802.3bs/bm give the lane requirements of 100G/200G/400G Ethernet (launch power, OMA, extinction ratio, SMSR, RIN, TDECQ, eye mask), which I use as a measurement template. ANSI Z136 limits the power from an open fibre end, so it caps the 16 mW total.

(f) System-level parameters for soliton / DSP compensation and integration

Soliton transmission depends on T_0, $|\beta_2|$, γ, the launch power and the amplifier spacing compared with L_D. Digital backpropagation depends on whether the receiver sees the complex field and how much computation six spans and eight channels need. Integration depends on footprint, power (mostly the TEC), coupling loss, reliability and scalability.

Problem Requirement Statement. The proposed transmitter must launch eight independent 25 Gb/s NRZ channels on the 100 GHz grid around 1550 nm, each from a single-mode, low-chirp source with SMSR ≥ 30 dB, a modulation bandwidth of at least 17.5 GHz, and a wavelength held to ±0.1 nm over a 40 °C ambient change. It must give about 2 mW (+3 dBm) per channel over six 80 km spans while keeping the nonlinear phase below about 1 rad, staying below the SBS and SRS thresholds, and meeting the eye-safety limit on the 16 mW total, in a compact, power-efficient implementation that can be checked against G.957 and IEEE 802.3bs/bm.

STAGE 2 – INVESTIGATE AND SELECT THE OPTICAL SOURCE / TRANSMITTER CONFIGURATION

The transmitter must provide an optical output suitable for high-speed, multi-span DWDM operation. Work backward from the system requirement to identify the source characteristics that must be satisfied.

Determine the parameters required to make this decision:

• Identify the recombination processes in the active region and the role of the energy bandgap. Using $E_g = 0.80$ eV, calculate the emission wavelength ($\lambda \approx 1.24/E_g$) and, using $\tau_r = 10$ ns and $\tau_{nr} = 40$ ns, the internal quantum efficiency.

• Compare LED, FP, DBR, DFB, VCSEL and QD lasers using output power, spectral width, modulation speed, efficiency, threshold behaviour and fiber-coupling suitability.

• Identify the threshold condition for laser oscillation and explain how linewidth and side-mode suppression affect an 8-channel DWDM link.

• Select the most suitable optical source and justify the selection using at least four technical parameters.

• Identify one technically credible alternative and state the condition under which it could become preferable.

SOLUTION – STAGE 2

(a) Recombination, wavelength and internal quantum efficiency

Injected electrons and holes recombine radiatively (spontaneous and, above threshold, stimulated emission) or non-radiatively (defects, Auger, interfaces); only the radiative part gives light. The photon energy is about E_g, so:

$$\lambda \approx 1.24 / E_g = 1.24 / 0.80 = 1.55\ \mu m = 1550\ nm$$

This is right in the required window. For the efficiency:

$$\eta_{int} = (1/\tau_r) / (1/\tau_r + 1/\tau_{nr}) = \tau_{nr} / (\tau_r + \tau_{nr}) = 40\ ns / (10\ ns + 40\ ns) = 0.80\ (80\ \%)$$

$$\text{Total carrier lifetime } \tau = \tau_r \tau_{nr} / (\tau_r + \tau_{nr}) = (10 \times 40)/50 = 8\ ns$$

The external differential efficiency is $\eta_d = \eta_s/(h\nu/q) = 0.3\ W/A \div 0.80\ V = 0.375$ (37.5 %), lower than η_{int} because of cavity losses.

(b) Comparison of the candidate sources

Source	Output power	Spectral width	Modulation speed	Efficiency	Threshold	Fibre coupling / suitability
LED	about 0.01–0.1 mW into SMF	50–100 nm	up to about 0.5–1 GHz	low	none	poor into SMF; far too wide for 0.8 nm slots
FP laser	1–10 mW	1–5 nm, several modes	10–20 GHz	good	10–30 mA	usable, but mode partition noise and the spectrum covers several channels
DBR laser	5–20 mW	single mode, MHz–10s of MHz, SMSR > 35 dB	up to about 10 GHz direct	moderate (passive sections)	20–40 mA	good; tunable, but mode hops and more control sections
DFB laser	5–20 mW	single mode, 1–10 MHz, SMSR 40–50 dB	10–25 GHz direct, 30 GHz+ with EAM	good	10–20 mA	very good; standard DWDM source
VCSEL (1550 nm)	0.5–2 mW	single mode possible, tens of MHz	up to about 25 GHz	high at low power	1–3 mA	easy coupling but low power for long haul
QD laser	a few mW	narrow, small α_H	10–25 GHz	good, temperature-insensitive	low	promising for integration; less mature at 1550 nm

(c) Threshold condition, linewidth and side-mode suppression

Laser oscillation starts when the round-trip gain equals the round-trip loss and the phase condition is satisfied:

$$\Gamma g_{th} = \alpha_{int} + \frac{1}{2L}\ln\left(\frac{1}{R_1 R_2}\right) \qquad \text{and} \qquad 2\beta L = 2\pi m$$

Above this point $P = \eta_s(I - I_{th})$.

In an 8-channel link the neighbour is only 0.8 nm away, so side modes land in the next channel and also cause mode-partition noise that grows with distance. SMSR of at least 30 dB (G.957 minimum) is needed, and a DFB gives 40–50 dB. A narrow linewidth reduces the dispersion penalty but lowers the SBS threshold (Stage 4).

(d) Selection

I select a DFB laser. The reasons, with numbers: (1) its wavelength is 1.24/0.80 = 1.55 μm, on the target; (2) η_int = 80 % and η_d = 37.5 %, with a low threshold of 12 mA; (3) it gives 8.4 mW at 40 mA (Stage 3), which leaves room for modulator and multiplexer losses; (4) it is single-mode with SMSR of about 40 dB and a MHz linewidth, against 1–5 nm for the FP laser, which is wider than the 0.8 nm channel spacing; (5) its modulation bandwidth of 20 GHz is above the 17.5 GHz needed; (6) its wavelength moves at 0.1 nm/°C (a Fabry–Perot laser moves roughly four times faster, because its gain peak shifts), which a TEC can control.

(e) Credible alternative

The DBR laser. It is also single-mode and tunable through its grating section, so it becomes preferable if the network needs a reconfigurable or spare transmitter that can serve any of the eight slots. The price is more complicated control (several currents, mode hops). A QD laser would be preferred in future if the transmitter must run uncooled.

Support. λ = 1550 nm and η_int = 80 % match the requirement, and 8.4 mW (9.24 dBm) is well above the +3 dBm per channel finally needed, so the modulator (5 dB) and multiplexer (3.5 dB) losses can be covered by a small booster gain of about 2.3 dB. The LED fails on power, linewidth and speed, the FP laser on spectral width and SMSR, and the VCSEL on power at 1550 nm.

STAGE 3 – DEVELOP AND EVALUATE THE MODULATION, CHIRP AND SPECTRAL-STABILITY STRATEGY

The selected source must support the required data rate and wavelength grid without introducing unacceptable modulation-related or temperature-related penalties.

You must:

• Calculate the output power at the stated bias current using P = η_s(I − I_th) and express it in dBm.

• Identify the bandwidth requirement for a 25 Gb/s NRZ data stream using B ≈ 0.7R_b, determine whether the candidate 20 GHz modulation bandwidth is sufficient, and state the available bandwidth margin.

• Write the carrier and photon rate equations, identify the relaxation-oscillation frequency, and explain how it differs in small-signal and large-signal modulation.

• Analyze direct modulation and external modulation with respect to chirp, bandwidth, dispersion interaction and transmission distance.

• Calculate the uncontrolled wavelength drift for a 40 °C ambient variation, compare it with the 0.8 nm channel spacing, and determine the temperature stability a TEC must maintain for ±0.1 nm. Compare TEC control with injection-current control for wavelength tuning.

SOLUTION – STAGE 3

(a) Output power

$P = \eta_s (I - I_{th}) = 0.3\ \text{mW/mA} \times (40\ \text{mA} - 12\ \text{mA}) = 0.3 \times 28 = 8.4\ \text{mW}$

$P(\text{dBm}) = 10 \log_{10}(8.4\ \text{mW} / 1\ \text{mW}) = 9.24\ \text{dBm}$

(b) Bandwidth requirement and margin

$B \approx 0.7\, R_b = 0.7 \times 25\ \text{Gb/s} = 17.5\ \text{GHz}$

$\text{Margin} = 20\ \text{GHz} - 17.5\ \text{GHz} = 2.5\ \text{GHz}$ (2.5/17.5 = 14.3 %)

So 20 GHz is sufficient on paper, but the margin is small; parasitics, temperature and large-signal operation all reduce the real bandwidth.

(c) Rate equations and relaxation oscillation

For a single-mode laser, with N the carrier density and S the photon density:

$$\frac{dN}{dt} = \frac{I}{qV} - \frac{N}{\tau_n} - v_g\, g(N)\, S$$

$$\frac{dS}{dt} = \Gamma\, v_g\, g(N)\, S - \frac{S}{\tau_p} + \Gamma\, \beta_{sp}\, \frac{N}{\tau_r}$$

Here $g(N) = a(N - N_{tr})$ is the gain, Γ the confinement factor, τ_p the photon lifetime, β_{sp} the spontaneous-emission factor and $1/\tau_n = 1/\tau_r + 1/\tau_{nr}$ (8 ns). Linearising gives the relaxation-oscillation frequency:

$$f_r = \frac{1}{2\pi} \sqrt{\left(v_g\, a\, S_0 / \tau_p \right)} \approx \frac{1}{2\pi} \sqrt{\left((I/I_{th} - 1) / (\tau_p\, \tau_n) \right)}$$

so f_r grows with $\sqrt{(I - I_{th})}$. Here $I/I_{th} = 40/12 = 3.33$. The 3 dB bandwidth is about 1.55 f_r, so the stated 20 GHz means $f_r \approx 13$ GHz.

In small-signal modulation the equations stay linear and the response is a second-order low-pass with a peak near f_r, independent of the data pattern. In large-signal modulation (NRZ swings the current over a large part of the bias) the laser shows a turn-on delay $t_d \approx \tau_n \ln[I_p/(I_p + I_b - I_{th})]$, ringing at f_r, pattern-dependent jitter and larger chirp, so the usable bandwidth is lower than the small-signal figure.

(d) Direct versus external modulation

Aspect	Direct modulation (DML)	External modulation (EAM / MZM)
Chirp	Large. $\Delta\nu(t) = (\alpha_H/4\pi)[(1/P)dP/dt + \kappa P]$, with α_H of about 3–6, so several GHz of frequency swing at 25 Gb/s	Small. EAM α is about 0–1 and a push-pull MZM can be about 0. The laser stays CW
Bandwidth	Limited by f_r and parasitics (20 GHz here, margin only 2.5 GHz)	Set by the modulator, 30 GHz or more is common; the laser needs no speed
Interaction with dispersion	The chirp adds to the dispersion broadening, so the dispersion penalty appears after a few km	Nearly transform-limited, so the dispersion limit is set only by the data spectrum
Extinction ratio / jitter	Limited; turn-on delay and ringing	High (8–12 dB), clean edges
Reach	Short reach, roughly under 10 km at 25 Gb/s on SMF	Metro and long haul when dispersion is compensated
Cost / power / loss	Simple, cheap, no insertion loss	Extra device, 3–6 dB insertion loss, bias control

Dispersion tolerance goes as $1/B^2$. Scaling from about 1000 ps/nm at 10 Gb/s gives roughly 160 ps/nm (about 9–10 km of SMF) at 25 Gb/s, while the link accumulates $17 \times 480 = 8160$ ps/nm, about 50 times more. So dispersion compensation is needed anyway, and chirp from a DML would make it less effective.

(e) Wavelength drift and temperature control

$$\Delta\lambda = (d\lambda/dT) \times \Delta T = 0.1 \text{ nm/°C} \times 40 \text{ °C} = 4.0 \text{ nm}$$

$$4.0 \text{ nm} / 0.8 \text{ nm} = 5 \text{ channel spacings}$$

Without control, a laser would drift across five neighbouring channel slots, collide with the other channels and fall outside the AWG passband. For ±0.1 nm:

$$\Delta T_{allowed} = \pm 0.1 \text{ nm} / (0.1 \text{ nm/°C}) = \pm 1.0 \text{ °C}$$

So the TEC must hold the laser within ±1 °C. I would design for ±0.1 °C, giving ±0.01 nm, ten times better than the tolerance.

Current tuning shifts the wavelength through heating and carrier density (about 0.01 nm/mA, an assumed typical value) and is fast, but its range is below about 1 nm and it also changes the output power (0.3 mW per mA), chirp and extinction ratio. TEC control is slow and uses extra power, but it covers several nm without disturbing the power or modulation. So the TEC does the main job, a wavelength locker closes the loop, and current tuning is at most a small trim.

Support. The CW laser gives 8.4 mW (9.24 dBm), enough to pay for a 5 dB modulator loss. The 2.5 GHz margin is too thin for direct modulation, while a 25–30 GHz external modulator removes the problem, and the 4.0 nm drift against 0.8 nm spacing shows the TEC is not optional.

STAGE 4 – INVESTIGATE AND CONTROL THE OPTICAL NONLINEARITIES

The 8-channel DWDM link may experience nonlinear performance degradation as optical launch power is increased over six spans. The design engineer must determine which effects are relevant and how they can be controlled.

Determine and discuss:

• Calculate the effective length $L_{eff} = (1 - e^{-\alpha L})/\alpha$ for an 80 km span and the SPM nonlinear phase shift $\varphi_{NL} = \gamma P L_{eff}$ per span and over six spans for 2 mW per channel.

• Explain how XPM and FWM depend on the 100 GHz channel spacing, fiber dispersion and launch power, and why they cause phase distortion and crosstalk.

• Calculate the SBS and SRS threshold powers using the given relations and compare them with the 2 mW per-channel launch power and the 16 mW total launch power.

• Compare SBS and SRS in terms of physical origin, gain bandwidth, power-transfer behaviour and effect on DWDM performance, and explain how source linewidth affects the SBS threshold.

• Check the 16 mW total launch power against the eye-safety (ANSI Z136) requirement, stating your assumed limit, and identify two practical methods for reducing nonlinear penalties without unnecessarily reducing the received optical signal level.

SOLUTION – STAGE 4

(a) Effective length and SPM phase shift

$\alpha = 0.2$ dB/km $\div$ 4.343 = 0.0461 km^{-1} ; $\alpha L = 0.0461 \times 80 = 3.684$

$L_{eff} = (1 - e^{-\alpha L})/\alpha = (1 - e^{-3.684})/0.0461 = (1 - 0.0251)/0.0461 = 21.17$ km

φ_{NL} (one span) $= \gamma P L_{eff} = 1.3$ W^{-1}km$^{-1} \times 0.002$ W $\times 21.17$ km $= 0.0551$ rad

φ_{NL} (six spans) $= 6 \times 0.0551 = 0.330$ rad

This is below the usual design guide of about 1 rad, so SPM alone is acceptable. For comparison, the worst-case XPM from the 7 other channels (equal power, co-polarised, no walk-off) is $2\gamma(7 \times 2 \text{ mW})L_{eff} = 0.77$ rad per span, 4.6 rad over six spans, but this is only an upper bound because of walk-off.

(b) XPM and FWM

XPM. The Kerr effect makes the phase of one channel depend on the power of the others, about $2\gamma L_{eff} \Sigma P_j(t)$, twice the SPM efficiency. The neighbours' bit patterns become phase noise, which dispersion converts into intensity noise. It grows linearly with launch power and channel count. Spacing and dispersion matter through walk-off: with D = 17 ps/(nm·km) adjacent channels drift $17 \times 0.8 = 13.6$ ps/km apart, so a 40 ps bit walks through its neighbour in about 2.9 km, far less than $L_{eff} = 21$ km, and the XPM averages out.

FWM. Channels at f_i, f_j and f_k create a wave at $f_i + f_j - f_k$. With equal spacing the products fall exactly on the channels, so the crosstalk is in-band; for N = 8 there are $N^2(N - 1)/2 = 224$ products. The efficiency depends on the phase mismatch $\Delta\beta = \beta_2(2\pi\Delta f)^2$:

$\beta_2 = D\lambda^2/(2\pi c) \approx 21.7\ \text{ps}^2/\text{km}$; $\Delta\beta = 21.7\times10^{-27}\ \text{s}^2/\text{m} \times (2\pi \times 100\ \text{GHz})^2 \approx 8.6\ \text{km}^{-1}$

$\eta_\text{FWM} \approx \alpha^2 / (\alpha^2 + \Delta\beta^2) = (0.0461)^2 / ((0.0461)^2 + (8.6)^2) \approx 3 \times 10^{-5}$ (about −45 dB)

So in standard SMF dispersion makes FWM weak at 100 GHz spacing, but it would be serious in dispersion-shifted fibre. FWM power goes as P^3, so doubling the power raises it by 9 dB.

(c) SBS and SRS thresholds

$P_\text{th(SBS)} \approx 21\ A_\text{eff} / (g_B\ L_\text{eff}) = 21 \times 80\times10^{-12}\ \text{m}^2 / (5\times10^{-11}\ \text{m/W} \times 21.17\times10^3\ \text{m})$

$= 1.68\times10^{-9} / 1.0585\times10^{-6} = 1.59\times10^{-3}\ \text{W} = 1.59\ \text{mW}$ (+2.0 dBm)

$P_\text{th(SRS)} \approx 16\ A_\text{eff} / (g_R\ L_\text{eff}) = 16 \times 80\times10^{-12} / (6\times10^{-14} \times 21.17\times10^3)$

$= 1.28\times10^{-9} / 1.270\times10^{-9} = 1.008\ \text{W} \approx 1.0\ \text{W}$ (+30 dBm)

Quantity	Threshold	Launch power	Result
SBS (per channel)	1.59 mW (+2.0 dBm)	2 mW per channel (+3.0 dBm)	Exceeded by 26 %, about 1.0 dB
SRS (total power)	1.0 W (+30 dBm)	16 mW total (+12 dBm)	63 times below threshold, about 18 dB margin

SBS acts on each channel separately because its gain bandwidth is so narrow, so 2 mW is compared with the threshold, not 16 mW. SBS is therefore the dominant impairment, and SRS is not a problem (the channels cover only 5.6 nm, far below the ~100 nm Raman shift).

(d) SBS versus SRS and the effect of linewidth

Aspect	SBS	SRS
Physical origin	Interaction with acoustic phonons (electrostriction)	Interaction with optical phonons (molecular vibration)
Gain bandwidth	Very narrow, about 20–100 MHz, Stokes shift about 11 GHz (0.09 nm)	Very broad, about 12–15 THz, peak shift about 13 THz (about 100 nm)
Gain coefficient	$g_B = 5\times10^{-11}$ m/W	$g_R = 6\times10^{-14}$ m/W (about 800 times smaller)
Direction and power transfer	Backward; the forward signal is converted into a reflected Stokes wave towards the transmitter, saturating the transmitted power and adding noise	Mainly forward in DWDM; power moves from shorter-wavelength channels to longer-wavelength ones
Effect on DWDM	Limits power per channel; no crosstalk between channels because the gain is narrower than the channel spacing	Gives channel-to-channel power tilt and crosstalk; matters for many channels over wide bands
Threshold here	1.59 mW per channel	about 1.0 W total

The formula threshold is for a very narrow source; in general $P_\text{th} \propto (1 + \Delta\nu_s/\Delta\nu_B)$ with $\Delta\nu_B$ about 20 MHz. A few-MHz DFB is the worst case, and spreading the linewidth to about

100 MHz would raise the threshold about 1 + 100/20 = 6 times (7.8 dB). This is an estimate, but it shows why a CW laser with an external modulator needs linewidth spreading.

(e) Eye safety and ways to reduce nonlinear penalties

$$P_{total} = 8 \times 2\ \text{mW} = 16\ \text{mW} = 10 \log_{10}(16) = 12.04\ \text{dBm}$$

My assumed limit is +10 dBm (10 mW) for an open fibre end at 1550 nm (a conservative Class 1 type value; to be confirmed against the standard edition). The 16 mW total is 6 mW (2.0 dB) above it, so it is not Class 1 as it stands. It can be treated as a higher hazard level with labelling, restricted access and automatic laser shutdown (ALS). The alternative is 8 × 1.25 mW = 10 mW, which also puts each channel below the SBS threshold but costs 10 log(2/1.25) = 2.0 dB of OSNR.

Two practical ways to cut the penalties without losing signal: (1) large effective-area fibre; since $\gamma \propto 1/A_{eff}$ and the thresholds $\propto A_{eff}$, going from 80 to about 110–125 μm^2 cuts φ_{NL} by 27–36 % and raises the thresholds by the same ratio. (2) A small low-frequency dither (or phase modulation) to spread the linewidth and raise the SBS threshold. Alternating polarisation on adjacent channels also helps.

Support. I keep 2 mW per channel to protect the OSNR, add a linewidth dither to raise the SBS threshold about six times, use per-span dispersion management against XPM and FWM, and add an ALS controller with a power tap. If the dither cannot be proven, the fallback is about 1.25 mW per channel (OSNR cost 2 dB).

STAGE 5 – EVALUATE SOLITON TRANSMISSION, DIGITAL BACKPROPAGATION AND PHOTONIC INTEGRATION

The final design must maintain acceptable signal quality while controlling dispersive and nonlinear impairments, meeting the applicable standards and considering practical implementation.

You must:

• Identify the physical balance between anomalous chromatic dispersion and Kerr nonlinearity that enables soliton transmission.

• Using $T_0 = 15$ ps and $|\beta_2| = 20\ ps^2/km$, calculate the dispersion length from $L_D = T_0^2/|\beta_2|$, and the fundamental-soliton peak power from $L_{NL} = L_D$.

• Using $L_{NL} = 1/(\gamma P_0)$ at the 2 mW per-channel launch power, calculate L_{NL} and the soliton order $N = \sqrt{(L_D/L_{NL})}$, and decide whether soliton transmission is feasible for this link.

• Explain how digital backpropagation can compensate dispersive and Kerr nonlinear impairments, identify one practical limitation, and state whether it is appropriate for the complete 8-channel link.

• Identify the transmitter parameters specified by ITU-T G.957 and IEEE 802.3bs/bm (100G/200G/400G Ethernet PHYs) and state how the proposed transmitter would be checked against them.

• Compare discrete, monolithic InP, and Si-photonics (hybrid-integrated laser) transmitter implementations in terms of footprint, power consumption, coupling, reliability and scalability, and analyze how integration influences overall system performance.

SOLUTION – STAGE 5

(a) Dispersion–nonlinearity balance

In anomalous dispersion ($\beta_2 < 0$, as in standard fibre at 1550 nm) dispersion broadens a pulse and chirps it, and SPM adds a chirp of the opposite kind. When they cancel the pulse keeps its sech shape: a soliton. The balance is L_D = L_NL, i.e. $P_0 = |\beta_2|/(\gamma T_0^2)$, with N = 1.

(b) Dispersion length and fundamental-soliton power

$L_D = T_0^2/|\beta_2| = (15\text{ ps})^2 / (20\text{ ps}^2/\text{km}) = 225/20 = 11.25\text{ km}$

For N = 1: $L_{NL} = L_D = 11.25$ km, so $P_0 = 1/(\gamma L_{NL}) = 1/(1.3 \times 11.25) = 0.0684$ W = 68.4 mW (+18.3 dBm)

(c) Soliton order at 2 mW and feasibility

$L_{NL} = 1/(\gamma P_0) = 1/(1.3\text{ W}^{-1}\text{km}^{-1} \times 0.002\text{ W}) = 384.6\text{ km}$

$N^2 = L_D/L_{NL} = 11.25/384.6 = 0.0293$; $N = 0.171$

N is far below 1, so the nonlinearity is too weak to balance the dispersion (L_D is about 34 times shorter than L_NL) and the pulses would simply disperse. A fundamental soliton needs 68.4 mW per channel, 34 times (15.3 dB) more than 2 mW. For eight channels this is 8 × 68.4 = 547 mW (+27.4 dBm), which breaks the eye-safety limit and is 43 times (16 dB) over the SBS threshold per channel. Also the 80 km amplifier spacing is much longer than the soliton period $z_0 = (\pi/2)L_D = 17.7$ km, and each span loses 16 dB. **Soliton transmission is not feasible for this link.**

(d) Digital backpropagation (DBP)

DBP takes the complex optical field and runs it backwards through a virtual fibre using split-step solutions of the inverse nonlinear Schrödinger equation, applying the inverse dispersion and then the inverse Kerr phase e^(−jγ|A|²L_eff) in many small steps, so dispersion and the nonlinear phase can be removed (XPM and FWM too if all channel fields are known).

The main limitation is that it needs the complex field, so a coherent receiver; a direct-detection NRZ link gives no phase. Removing XPM and FWM would need all eight channels processed together (200 GHz or more of digitised bandwidth) with many steps over six spans, which is impractical. Single-channel DBP removes only SPM and not the signal–noise interaction or PMD. So DBP is **not appropriate for the complete 8-channel link**; with SPM of only 0.33 rad, ordinary dispersion compensation is enough.

(e) Standards parameters and how the transmitter is checked

G.957 specifies mean launched power, centre wavelength range, spectral width, minimum SMSR (30 dB), minimum extinction ratio and the eye mask. IEEE 802.3bs/bm specify per lane the wavelength range, average power, outer OMA, extinction ratio, SMSR, RIN, TDP/TDECQ and the eye mask. These Ethernet PHYs are for reaches of 10 km or less, so for the 480 km line I use them as the measurement template, with G.694.1/G.959.1 for the grid and line interface.

Parameter	Proposed value	How it is checked
Centre wavelength vs grid	on 100 GHz grid, within ±0.1 nm	Optical spectrum analyser or wavemeter, over a 40 °C chamber test
SMSR and spectral width	≥ 30 dB (design 40 dB); MHz linewidth	OSA for SMSR; delayed self-heterodyne for linewidth
Average power per channel / total	+3 dBm per channel, +12 dBm total	Optical power meter at the output; compare with the hazard-level limit
Extinction ratio and eye	ER ≥ 8 dB (assumed), eye mask passed with margin	Sampling oscilloscope with a 25 Gb/s reference receiver
Dispersion penalty / BER	within the standard's penalty limit	BER tester with PRBS31 through the compensated link
RIN	within the standard's limit	RIN measurement setup
Eye safety / ALS	shutdown on fibre break	Fibre-break test and shutdown time measurement

(f) Discrete, monolithic InP and Si-photonics implementations

Aspect	Discrete (8 modules + external AWG)	Monolithic InP PIC	Si photonics + hybrid laser
Footprint	Large, a full line-card area	Very small, a few mm² chip	Smallest at volume
Power consumption	Highest: eight TECs, eight drivers	Low: shared TEC, short interconnects	Low, but laser still needs cooling; heaters for tuning the AWG
Coupling	Many fibre and lens interfaces, about 0.5–1.5 dB each	Only the output fibre; laser, modulator and mux are on one chip	Laser-to-chip coupling about 1.5–3 dB plus Si modulator loss of 4–6 dB
Reliability	Many joints and alignments, but each part proven	Fewer joints, mature InP lasers	Depends on the III-V bonding; Si is thermally sensitive
Scalability	Poor, cost and size grow with N	Good up to about 8–16 channels; yield falls as complexity grows	Best for high volume and many channels
Effect on system performance	Best parts but highest loss and drift	Best laser quality, low insertion loss, low chirp	Extra loss lowers launch power and OSNR; Si and two-photon absorption limit high power

Integration affects the system mainly through loss, temperature and parasitics: every dB of coupling or insertion loss is a dB less OSNR. One chip puts lasers, modulators and mux at the same temperature, so one TEC and locker serve all eight channels, and short on-chip paths ease the modulator bandwidth.

Support. DFB source with 80 % η_int and 8.4 mW; the EAM array removes the chirp and margin problems; the SPM budget is small and SBS is controlled by dither, so no nonlinear compensation is needed in the transmitter (single-channel DBP could be added in a future coherent receiver); solitons are rejected (N = 0.17); and the InP chip saves seven TECs and most of the coupling loss compared with a discrete build, and avoids the laser-bonding and modulator loss of Si photonics.

STAGE 6 – VERIFY AND CRITICALLY EVALUATE THE COMPLETE TRANSMITTER SOLUTION

Prepare a final verification table using the results obtained from Stages 2–5.

Requirement	Specified / Target	Achieved	Satisfactory?
Optical source selection	1550 nm, single-mode, narrow linewidth, good efficiency	DFB: λ = 1.24/0.80 = 1.55 μm; η_int = 80 %; SMSR ≥ 30 dB (typ. 40 dB); linewidth in MHz	Yes
Modulation bandwidth and output power	≥ 17.5 GHz; enough power for +3 dBm per channel after losses	20 GHz (margin 2.5 GHz, 14.3 %); P = 8.4 mW = 9.24 dBm; booster gain about 2.3 dB gives +3.0 dBm/ch	Yes (external modulator used because the margin is thin)
Spectral/chirp and wavelength stability	±0.1 nm over 40 °C; low chirp	Uncontrolled drift 4.0 nm (5 spacings); TEC ±0.1 °C gives ±0.01 nm (limit ±1 °C); EAM/MZM gives low chirp	Yes, with TEC and locker
Nonlinear-impairment control and eye safety	φ_NL below about 1 rad; below SBS/SRS thresholds; total power within eye-safety limit	φ_NL = 0.055 rad/span, 0.33 rad total; SRS 1.0 W vs 16 mW; SBS 1.59 mW vs 2 mW (exceeded); 12 dBm vs assumed 10 dBm limit	Partly: met only with SBS dither and ALS (or 1.25 mW per channel)
Soliton / dispersion condition	N close to 1 if solitons are to be used	L_D = 11.25 km; P_0 = 68.4 mW; L_NL = 384.6 km at 2 mW; N = 0.17	No for solitons (rejected); dispersion handled by compensation instead
Integration / DSP / standards strategy	Meets G.957 and IEEE 802.3bs/bm parameters; compact; compensation approach chosen	Monolithic InP PIC + booster; per-span dispersion compensation; DBP not used on the full link; parameters checked by measurement plan	Yes (to be confirmed by measurement)

Your verification must explain whether the calculated and analyzed results are consistent with the original communication requirement.

Consistency with the requirement. Most results agree: 1550 nm with 80 % efficiency, enough bandwidth (using an external modulator), wavelength held to ±0.1 nm, small SPM phase, and standards parameters that can be met. Two results do not match the nominal numbers: at 2 mW the SBS threshold (1.59 mW) is exceeded, and the 16 mW total is above my assumed +10 dBm eye-safety limit. The design is consistent only with the dither and ALS included, otherwise the power must drop to about 1.25 mW per channel. The soliton check fails (N = 0.17), so the design does not rely on solitons. All results are calculated, so a prototype must be measured before sign-off.

FINAL TRANSMITTER DESIGN – INTEGRATED ENGINEERING SOLUTION

Create the final transmitter configuration rather than merely selecting an existing option. Explain how the individual design choices work together to satisfy the 8-channel DWDM communication requirement.

The final solution must include:

• a final parameter/design table showing the adopted optical source and transmitter characteristics (wavelength, power, bias, bandwidth, TEC setting);

• a clearly labelled block diagram of the proposed DWDM transmitter;

• the selected modulation and wavelength-control approach with its bandwidth and drift justification;

• the selected nonlinear-control, eye-safety and compensation strategy;

• the selected photonic-integration/discrete implementation approach and the applicable standards;

• the calculated dispersion length, soliton order and the stated soliton feasibility;

• a final engineering recommendation supported by the preceding stages.

SOLUTION – FINAL TRANSMITTER DESIGN

Final parameter / design table

Item	Adopted value
Channels and grid	8 channels, 100 GHz (0.8 nm) spacing, 193.1–193.8 THz (about 1552.5–1547.0 nm), ITU-T G.694.1 grid (assumed)
Optical source	InGaAsP/InP single-mode DFB laser per channel, CW, externally modulated
Emission wavelength	$\lambda = 1.24/E_g = 1.24/0.80 = 1.55$ μm
Efficiency	$\eta_{int} = 80$ % ($\tau = 8$ ns); $\eta_d = 37.5$ %
Threshold / bias current	$I_{th} = 12$ mA ; $I = 40$ mA ($I/I_{th} = 3.33$)
Output power per laser	$P = 0.3 \times (40 - 12) = 8.4$ mW = 9.24 dBm

Item	Adopted value
Modulation and bandwidth	25 Gb/s NRZ, EAM or MZM (assumed ≥ 25 GHz); required 0.7 × 25 = 17.5 GHz; laser 20 GHz, margin 2.5 GHz
TEC setting	25 °C set point (assumed), stability ±0.1 °C (maximum allowed ±1 °C), closed loop with wavelength locker; drift 0.1 nm/°C × 0.1 °C = ±0.01 nm
Launch power	2 mW (+3.0 dBm) per channel; 16 mW (+12 dBm) total, after booster of about 2.3 dB gain
Linewidth / SMSR	MHz-level DFB linewidth, spread by dither for SBS; SMSR ≥ 30 dB (typically 40 dB)
Nonlinear results	L_eff = 21.17 km; φ_NL = 0.055 rad/span, 0.33 rad for six spans; P_th(SBS) = 1.59 mW; P_th(SRS) = 1.0 W
Soliton check	L_D = 11.25 km; P_0 = 68.4 mW; L_NL = 384.6 km at 2 mW; N = 0.17, not feasible
Compensation	Per-span dispersion compensation and linear electronic compensation at the receiver; no soliton, no full-link DBP
Implementation	Monolithic InP PIC (8 DFB + EAM array + AWG), single package with TEC, locker, booster and ALS tap
Standards	ITU-T G.957 (and G.694.1 / G.959.1), IEEE 802.3bs/bm parameter templates, ANSI Z136 eye safety

Block diagram

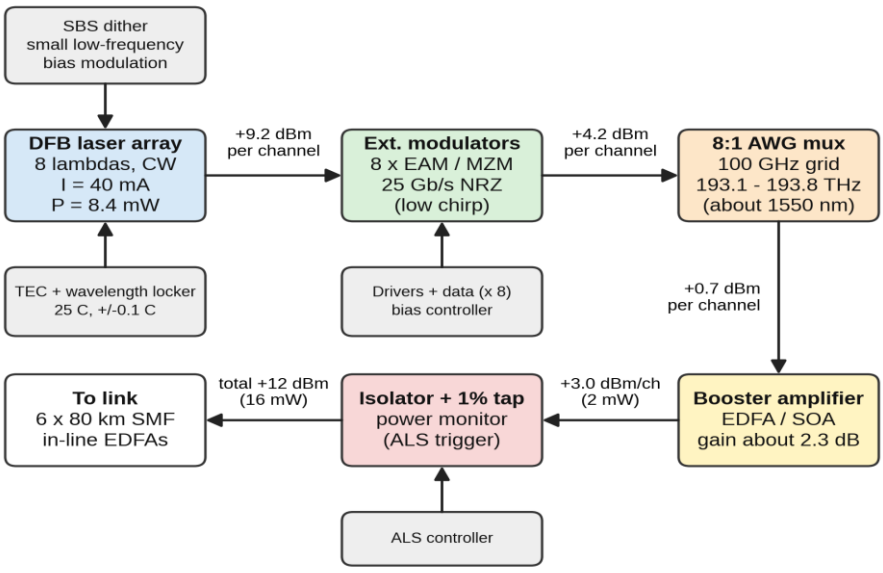


Figure 1. Block diagram of the proposed 8-channel DWDM transmitter. Power levels are per channel and use the assumed losses of 5 dB (modulator) and 3.5 dB (AWG): 9.24 → 4.24 → 0.74 dBm, then the booster brings each channel to +3.0 dBm (2 mW) and the total to +12 dBm (16 mW).

How the choices work together

Modulation and wavelength control. The DFB lasers run CW at 40 mA and external modulators put on the data, avoiding chirp and the thin 2.5 GHz margin. The uncontrolled drift of 4.0 nm (five spacings) is held to about ±0.01 nm by a TEC at 25 °C (±0.1 °C) and a wavelength locker, well inside ±0.1 nm.

Nonlinear control, eye safety and compensation. 2 mW per channel gives only 0.33 rad of SPM over six spans, and FWM and SRS are negligible. SBS (1.59 mW threshold) is handled by a small low-frequency dither that raises the threshold about six times. The 12 dBm total exceeds my assumed +10 dBm limit, so the output has a power tap and an ALS controller. Dispersion is compensated per span; DBP is not used.

Final recommendation. Build the transmitter as an 8-channel InP DFB + EAM photonic-integrated chip with TEC and wavelength locker, SBS dither, a booster giving 2 mW per channel, ALS protection and per-span dispersion compensation. It is checked against G.957 and IEEE 802.3bs/bm parameters as listed in Stage 5. Soliton transmission is not used (L_D = 11.25 km, N = 0.17).

FINAL ENGINEERING DECISION

WOULD YOU DEPLOY THE TRANSMITTER SOLUTION YOU CREATED?

State clearly:

YES / NO – WITH TECHNICAL JUSTIFICATION

Your decision must be supported by the source-selection, modulation-bandwidth, wavelength-stability, nonlinear-impairment, soliton/compensation, standards and integration analysis.

YES – I would deploy this transmitter, with two conditions. (1) The SBS dither must be fitted and shown in testing to raise the threshold above 2 mW, and (2) the ALS and hazard-level labelling must be in place for the 12 dBm output. If either condition cannot be met, I would reduce the launch power to about 1.25 mW per channel (10 mW total) instead of deploying at 2 mW.

Justification. The DFB source gives 1550 nm, 80 % internal efficiency and 8.4 mW. The 20 GHz laser bandwidth is only 2.5 GHz (14 %) above the 17.5 GHz needed and direct modulation would chirp, so external modulation is right. The 4.0 nm uncontrolled drift is held within ±0.1 nm by the TEC. The SPM phase of 0.33 rad is acceptable, SRS (1.0 W) and FWM (about −45 dB) are not a problem, and the one threshold that is crossed, SBS, is handled by dither. Solitons are impossible (N = 0.17) and DBP is impractical for eight channels, so dispersion compensation is used. InP integration gives the lowest coupling loss, footprint and power. The remaining risk is that everything is calculated, not measured, so a prototype must be tested for SBS, eye mask and wavelength drift before volume deployment.

SUBMISSION REQUIREMENTS

• The assignment shall be submitted as one structured engineering solution.

• All calculations must show equations, substitutions, units and intermediate steps.

• Source-selection and implementation decisions must be supported by technical reasoning.

• Where comparison is required, present the comparison in a clear table.

• Include the final transmitter block diagram and final design/parameter table.

• State assumptions wherever a parameter is not explicitly given.

• Equivalent technically valid approaches may be accepted when supported by appropriate engineering reasoning.

SCHEME OF EVALUATION

Criterion	**Excellent**	**Good**	**Satisfactory**	**Needs Improvement**
Problem-solving accuracy	Correct principles, calculations and units throughout.	Minor errors with correct overall approach.	Partially correct calculations or concepts.	Major errors or incomplete solution.
Analytical reasoning	Strong analysis of source, modulation and nonlinear effects.	Good analysis with relevant technical reasoning.	Basic analysis with limited depth.	Mostly descriptive or unsupported.
Engineering recommendation	Integrated and feasible recommendation strongly supported by results.	Suitable recommendation with reasonable support.	General recommendation with partial support.	Recommendation inconsistent with analysis.
Technical presentation	Clear, structured and professionally presented.	Mostly clear and organized.	Adequate but uneven.	Unclear or poorly structured.

PERFORMANCE EXPECTATIONS

Excellent

The student systematically evaluates the communication requirement, compares optical-source and modulation alternatives, performs accurate calculations, analyzes nonlinear and soliton conditions, makes technically defensible decisions, and provides a coherent final transmitter recommendation.

Good

The student demonstrates correct problem-solving methodology and appropriate engineering decisions, with minor errors or limited justification in some areas.

Satisfactory

The student obtains a workable transmitter configuration but relies mainly on standard calculations with limited comparison, justification or critical evaluation.

Needs Improvement

The student demonstrates incomplete problem solving, significant calculation or conceptual errors, weak engineering justification, or an inconsistent final recommendation.